

LESSON 321 (1.1 DUAL, 0.9 INSTRUMENT)

Lesson Objective

During this lesson, the student will increase proficiency in precision and non-precision instrument approach procedures. GPS approaches will be introduced.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

Lesson Content (Ground)

Go/no-go Decision

- The student must make a go/no-go decision depending on:
 - Personal minimums.
 - FIT minimums.
 - Aircraft.
 - Airspace.
- Review the student's planning with a focus on:
 - Weather, weight and balance, performance, flight planning, and fuel calculations.
 - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
 - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
 - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

Lesson Content (Flight)

Aircraft Checklist Items

Emphasize timeliness to minimize ground time!

- Instrument clearance
 - Instructors, you could either file to pick up your clearance on the ground/in the air OR pick up practice VFR approaches in the air.
 - Make sure your student is NOT the one filing.
 - File KMLB to KMLB and choose an IAF or FAF of your choosing from the approaches in KMLB.
 - In the remarks section, list the approaches desired or "multiple practice approaches."
- Instrument Cockpit Check
 - Student Tells, Student Does.
 - Have the student read each item from the checklist out loud.

- Discipline the student to set bugs, bearing pointers, and any auxiliary features.
- The autopilot check can be conducted now:
 - Engage autopilot.
 - Test roll: left and right.
 - Test pitch: up and down (VS mode).
 - Ensure autopilot can be overridden.
 - Check the red disconnect button.
- Electric trim test:
 - Check the motion of the trim wheel.
 - Check the S-TEC trim annunciators.
 - Each half-switch should not move the trim wheel.
 - The disconnect switch should stop the wheel.
- Before taxi check:
 - Student Tells, Student Does.
 - Airspeed indicator (PFD and standby): zero.
 - Attitude indicator (PFD): pitch/bank deviations +/- 5 degrees.
 - Standby attitude indicator: standby battery check and adjust the miniature aircraft.
 - Altimeter (PFD and standby): +/- 75 feet of station elevation.
 - Vertical speed indicator (PFD): reads 0 or use current indication.
 - Trim coordinator: wings level or no vector when still; inclinometer centered when still; skids when turning.
 - HSI: matches compass and turns appropriately.
 - Magnetic compass: indicates heading properly, no bubbles, card present, and swings freely.

Departure

- Follow the tower's instructions for departure.
- The student must put the hood or foggles on when safe.
- Climb out:
 - Student Does, Instructor Evaluates.
 - Maintain V_Y .
 - Conduct the 1,000ft check.
 - Level off at a given altitude.
 - Contact departure when handed over by tower. Check in with only heading and altitude initially.

Instrument Approach Procedures

ILS

- Tell the student to expect an ILS approach.
 - Nearby options include the ILS 09R @ KMLB, the ILS 36 @ KTIK, or the ILS 03/21 @ KCOF.

- Consider engaging the autopilot while the student completes all the steps of the **WIRE** checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS)
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
 - **Radios:**
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course.
 - **Enroute brief**
 - Emphasize the importance of this being a brief and not explaining the plate. Students will tend to miss radio calls as they will be talking over approach. Instructors, make sure you are paying attention to the frequency and also listening to your student briefing the plate. Adjust the comm volumes accordingly.
- Prior to localizer intercept and when cleared for the approach, ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
- Remind them that setting around 1,800RPM will provide a roughly -500fpm descent rate at 90 knots.
- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Ensure the student executes their missed approach upon reaching the DA:
 - Set full power and pitch up for Vy
 - Upon reaching a positive rate of climb, retract the flaps
 - Turn to the missed approach heading/radial
 - Notify ATC

GPS approach

- Ask ATC to begin a GPS approach from an IAF (for example, NAJBE for the RNAV 09R @ KMLB). If given a direct to fix instruction, proceed to this fix via the PROC page.
- Ensure the student completes all the steps of the **WIRE** checklist:
 - **Weather**
 - **Instruments:**
 - Set both altimeters.
 - Set the correct nav source on the PFD (VLOC/GPS).
 - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the

approach will begin from radar vectors.

- Radios:
 - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
 - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
 - Set the final approach course if required.
- Enroute brief
- Point out the G430 prompts to turn just prior to reaching fixes on the approach so as not to overshoot.
 - If autopilot is to be used, engage GPSS mode by pressing NAV twice to ensure this happens automatically.
- Prior to final approach course intercept and when cleared for the approach, ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
- At the FAF, remind the student that setting around 1,800RPM and pitching for 90 knots will provide a -500fpm descent rate to track the glideslope.
- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Remind the student of the 91.175(c) criteria required to land. Upon reaching the DA, have the student go visual and land.

Post-Brief

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 322.
- Assign Homework:
 - Continue studying instrument approach procedures.

Completion Standards

The student shall demonstrate complete understanding of all appropriate precision and non-precision approach procedures.